LIBRARY MANAGEMENT SYSTEM

Using MongoDB

A Database Management Project

Submitted By	Shalini Maurya
Roll No.	1250258412
Course	BCA (Data Science & AI)
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University	BBD University
Teacher	Harendra Singh
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DECLARATION

I hereby declare that this project report is prepared for academic purposes and presents a Library Management System implemented using MongoDB concepts and commands.

1. INTRODUCTION

A Library Management System is a database application used to manage books, students or members, and book issue/return information. In this project, MongoDB is used as the database because it is a NoSQL, document-oriented database.

2. OBJECTIVES

• Store library book records.
• Store student/member details.
• Record book issue and return information.
• Search books quickly.
• Update and delete records when required.
• Reduce manual paperwork.

3. TECHNOLOGIES USED

Database: MongoDB
Database Type: NoSQL
Query Language: MongoDB Query Language
Tools: MongoDB Compass / MongoDB Shell

4. SYSTEM REQUIREMENTS

Hardware: Basic computer/laptop with sufficient storage and RAM.
Software: MongoDB Community Server, MongoDB Compass or MongoDB Shell.

5. DATABASE DESIGN

The project uses three main collections:

Collection	Purpose
books	Stores book information
students	Stores student/member information
issue_books	Stores issue and return transactions

6. Create Database

```
use LibraryManagement
```

7. Insert Book Records

```
db.books.insertMany([
  { book_id: 101, title: "Python Programming", author: "John Smith", category: "Programming"
, quantity: 5 },
  { book_id: 102, title: "Database Management System", author: "Raghu Ramakrishnan", categor
y: "Database", quantity: 3 },
  { book_id: 103, title: "Computer Networks", author: "Andrew S. Tanenbaum", category: "Netw
orking", quantity: 4 }
])
```

8. Insert Student Records

```
db.students.insertMany([
  { student_id: 1, name: "Shalini", course: "BCA", semester: 2 },
  { student_id: 2, name: "Priya", course: "BCA", semester: 2 }
])
```

9. Issue a Book

```
db.issue_books.insertOne({
  student_id: 1,
  book_id: 101,
  issue_date: "2026-10-06",
  return_date: null,
  status: "Issued"
})
```

10. Display All Books

```
db.books.find()
```

11. Search a Book

```
db.books.find({ title: "Python Programming" })
```

12. Update Book Quantity

```
db.books.updateOne(
  { book_id: 101 },
  { $set: { quantity: 4 } }
)
```

13. Return a Book

```
db.issue_books.updateOne(
  { student_id: 1, book_id: 101 },
  { $set: { return_date: "2026-10-10", status: "Returned" } }
)
```

14. Delete a Book

```
db.books.deleteOne({ book_id: 103 })
```

15. CRUD OPERATIONS

Create: insertOne() and insertMany() add records.
Read: find() retrieves records.
Update: updateOne() changes existing records.
Delete: deleteOne() removes records.

16. SAMPLE OUTPUT / SCREENSHOT SECTION

Run the above commands in MongoDB Compass or MongoDB Shell and paste screenshots of the following outputs:

• Database and collections

• Inserted book records

• Student records

• Book issue record

• Search / Find result

• Update and return result

17. ADVANTAGES

• Easy to store and retrieve library data.
• Flexible document structure.
• Fast searching and updating.
• Less manual work.
• Easy to extend with additional collections.

18. LIMITATIONS

• A real library system may require authentication and role management.
• Backup and security should be configured for production use.
• A complete application would also need a user interface.

19. FUTURE SCOPE

The system can be extended by adding a web/mobile interface, login system, fine calculation, book availability notifications, barcode scanning, and an admin dashboard.

20. CONCLUSION

The Library Management System using MongoDB demonstrates how a NoSQL database can be used to manage books, students, and issue/return transactions. The project covers the basic CRUD operations and provides a simple foundation for developing a complete library application.

— END OF PROJECT —